LESSON 11 · CHEAT SHEET

SQL I: Asking the Vault

FROM PYTHON

```
import sqlite3
con = sqlite3.connect("data/guild.db")
rows = con.execute("""
    SELECT name, level FROM heroes
""").fetchall()         # list of tuples
for name, level in rows:
    print(f"{name:<20}{level:>3}")
n = con.execute("SELECT COUNT(*) FROM heroes"
                ).fetchone()[0]  # one value
```

THE ORDER (ALWAYS)

```
SELECT   columns, COUNT(*) AS n
FROM     table
WHERE    row filter
GROUP BY columns
HAVING   group filter
ORDER BY column DESC, other
LIMIT    5
```

WHERE

```
WHERE class = 'Ranger'     # exact text
WHERE level >= 12 AND active = 'yes'
WHERE outcome <> 'Success'  # not equal
WHERE (a = 1 OR b = 2) AND c = 3
WHERE kin IN ('Dwarf', 'Gnome')
WHERE level BETWEEN 12 AND 20 # inclusive
WHERE title LIKE '%mill%'  # % = any text
WHERE region IS NULL       # never = NULL
WHERE NOT outcome IN ('Fled')
```

SORT, CUT, DISTINCT

```
ORDER BY reward DESC       # big first
ORDER BY days, reward DESC # two levels
LIMIT 5                    # top 5
SELECT DISTINCT monster    # unique
SELECT reward / 100 AS hundreds
```

AGGREGATES

```
COUNT(*)   COUNT(DISTINCT region)
SUM(paid)  AVG(reward)  MIN(days)  MAX(days)
ROUND(AVG(reward), 2)
SUM(paid) * 1.0 / SUM(days)  # keep decimals
```

GROUP BY + HAVING

```
SELECT monster, COUNT(*) AS n,
       ROUND(AVG(reward)) AS avg
FROM quests
WHERE outcome = 'Success'   # rows first
GROUP BY monster
HAVING n >= 30               # then groups
ORDER BY avg DESC
```

EXCEL → PANDAS → SQL

EXCEL	PANDAS	SQL
AutoFilter	`df[df["level"] >= 12]`	`WHERE`
Hide columns	`df[["a", "b"]]`	`SELECT a, b`
Sort Z–A	`ascending=False`	`DESC`
Top 5	`.head(5)`	`LIMIT 5`
Remove dupes	`.unique()`	`DISTINCT`
COUNTA	`len(df)`	`COUNT(*)`
PivotTable	`.groupby("m")`	`GROUP BY`
Filter a pivot	`t[t["n"] >= 30]`	`HAVING`

GOTCHAS

- Text in 'single quotes', exact case
- No comma between columns: the 2nd becomes an alias
- WHERE can't use COUNT/AVG: that's HAVING
- int / int drops decimals: `* 1.0`
- ASC is the default: say DESC for "biggest"
- AND before OR: use brackets